

Manufacturing Automation
An overview of Mechatronics

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Vertical Machining Centre

Hydraulics system

1. Pump
2. Actuator
3. Hydraulic oil
4. Control devices
5. Filters

Painting

1. Green paint

Spindle System

1. Housing
2. Spindle
3. Spindle Motor
4. Transmission
5. Bearings
6. Tool clamping

Coolant System

1. Coolant
2. Coolant pump

Machine Structure

1. Bed
2. Column
3. Slides

Lubrication System

1. Lubricant
2. Lubrication pump
3. Filters

Feed Drive System

1. Feed motors
2. Ball screws
3. Ball screw housing
4. Transmission
5. Guide ways
6. Bearings
7. Table
8. Telescopic covers

Feed back system

1. Encoder / Scales
2. Sensors

Tooling

1. Inserts
2. Holders

Enclosures

1. Cabinet
2. Guard

Tool Changer

1. ATC servo motor
2. Bearings
3. Indexing mechanism
4. Drum

Lighting

1. Machine lamp
2. Cabinet lamp
3. Tower lamp

Control System

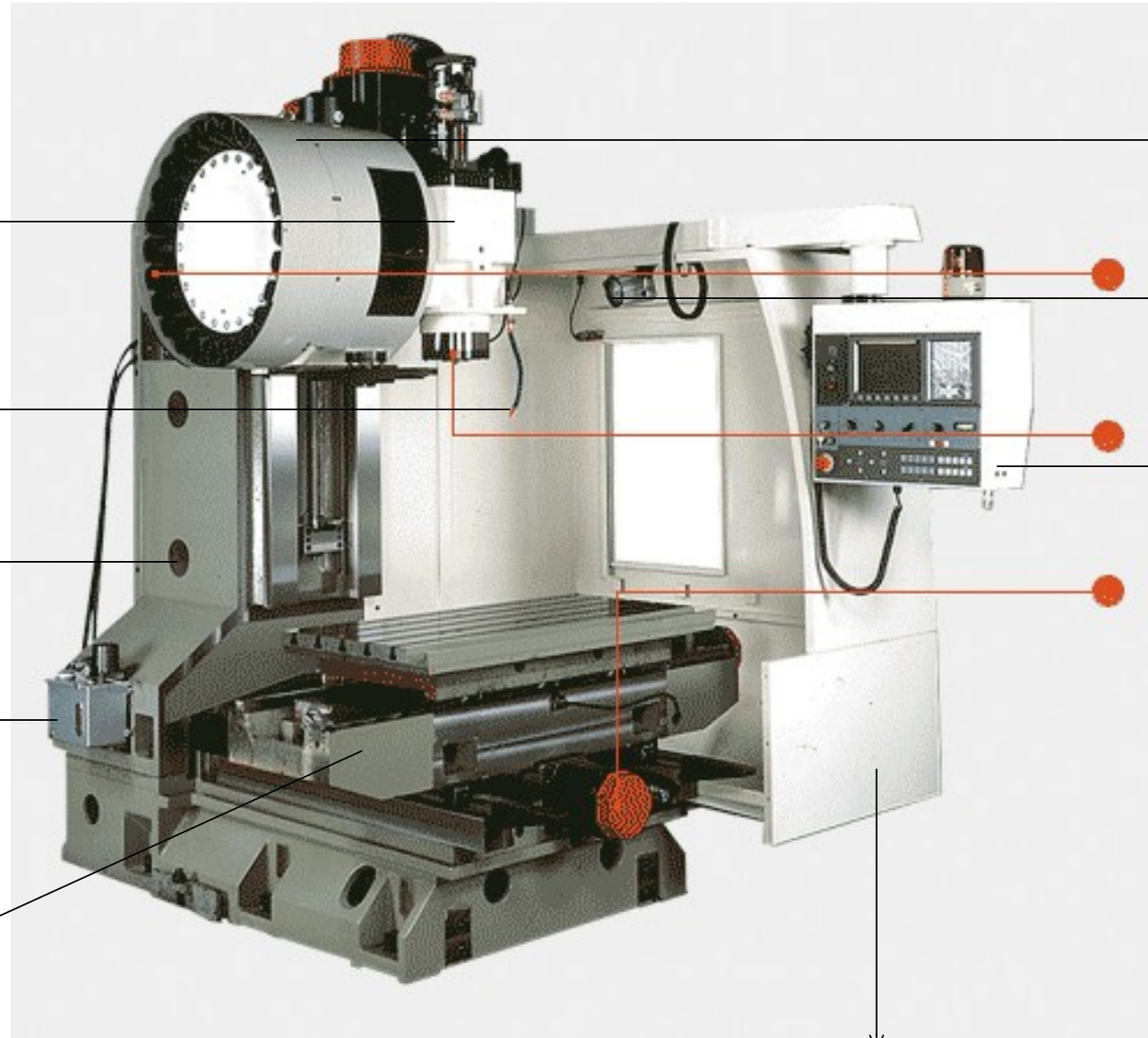
1. Monitor
2. Operating panel

Electrical system

1. Transformer
2. Cooling fan
3. 24V power supply
4. Drives
5. Air conditioner
6. Cabinet cooling

Chip Conveyor

1. Conveyor motor
2. Mechanism



General principles of Automation

- *The automation of industry can manifest itself in a variety of forms, depending on the type of industry and its production requirements.*
- *The primary aim of automation is to first help mankind, then perhaps eventually replace men who work at tiring or repetitive tasks.*

- *This mechanization helps obtain economic production with more uniform quality.*
- *The production process and the design of products is often modified to make manufacturable by automatic production process.*

Stages of *Automation*

- *The first stage in the process is to replace muscle power with mechanical power, which is controlled directly by a worker. The worker can then apply his energies to increasing productivity and production efficiency.*
- *The second stage is to co-ordinate automatically the movements of various power devices with control systems. This method still requires to feed manually and remove the component parts and to check the production quality.*

- *The final stage is the automation of component part feeding and removal, along with automatic quality inspection. At this stage the machine is completely automated- the workers mental and physical energies are no longer directly required in the manufacturing process. The worker will be needed only to monitor the machine's operation and to make adjustments or repairs when there is a failure.*

The Control Power Dialogue

A typical machine consists of the following:

- *The Power system. In the example it is a drill head mounted on a cylinder to provide feeding action.*
- *The Control system. This co-ordinates the automatic cycle of the machines power system and allows for manual intervention by the operator*
- *The control system also has pre-defined interlocks*

The control system also includes:

- *The input devices, consisting of the required sensing devices.*
- *The Logic circuits, which contains the program instructions.*
- *The manual control panel for operator intervention.*
- *To ensure proper operation and to protect the machine from damaging itself or the operator when a malfunction occurs, a complete dialogue is usually maintained between the power system and the control system.*

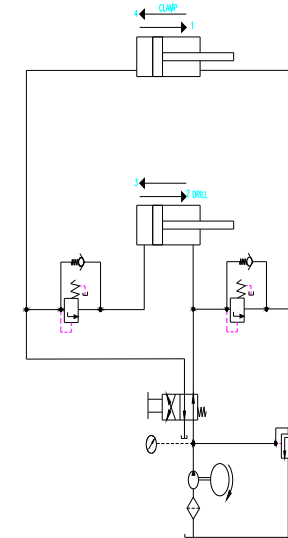
❖ *THIS FIG. SHOWS A TYPICAL APPLICATION OF A SEQUENCE VALVE TO ENSURE THE CLAMPING OF THE WORKPIECE PRIOR TO THE COMMENCEMENT OF DRILLING*

CLAMP

❖ *FURTHER THE RETRACTION OF THE DRILL PRIOR TO THE DECLAMPING OF THE WORKPIECE.*

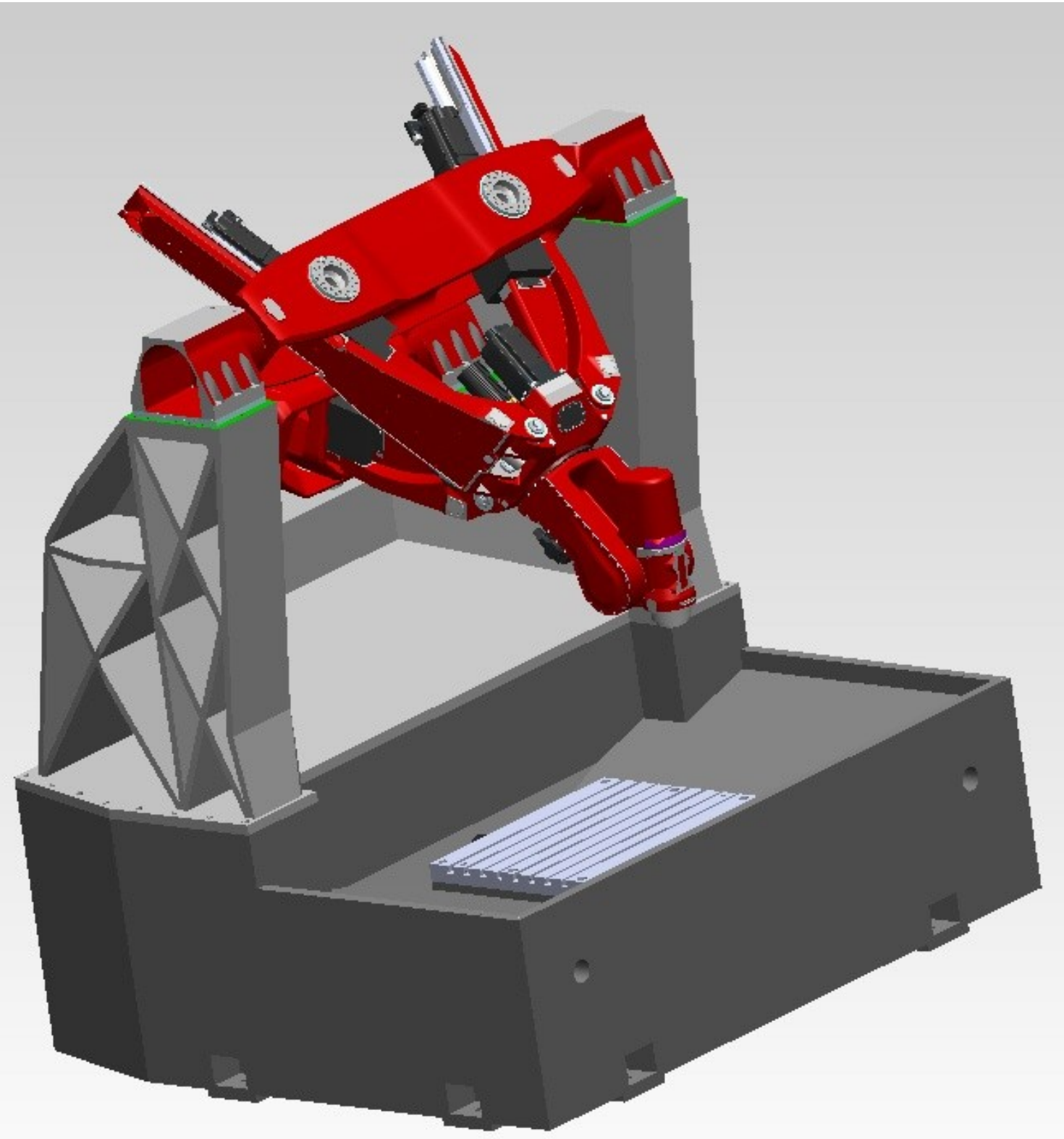
❖ *THE SEQUENCE VALVES ARE SET TO SPECIFIC PRESSURE LEVELS SO AS TO ENSURE THE PRIORITIES OF FUNCTIONS.*

❖ *THE RETURN FLOW FROM THE 2 CYLINDERS IS EFFECTED BY THE CHECK VALVES IN PARALLEL WITH THE SEQUENCE VALVES.*



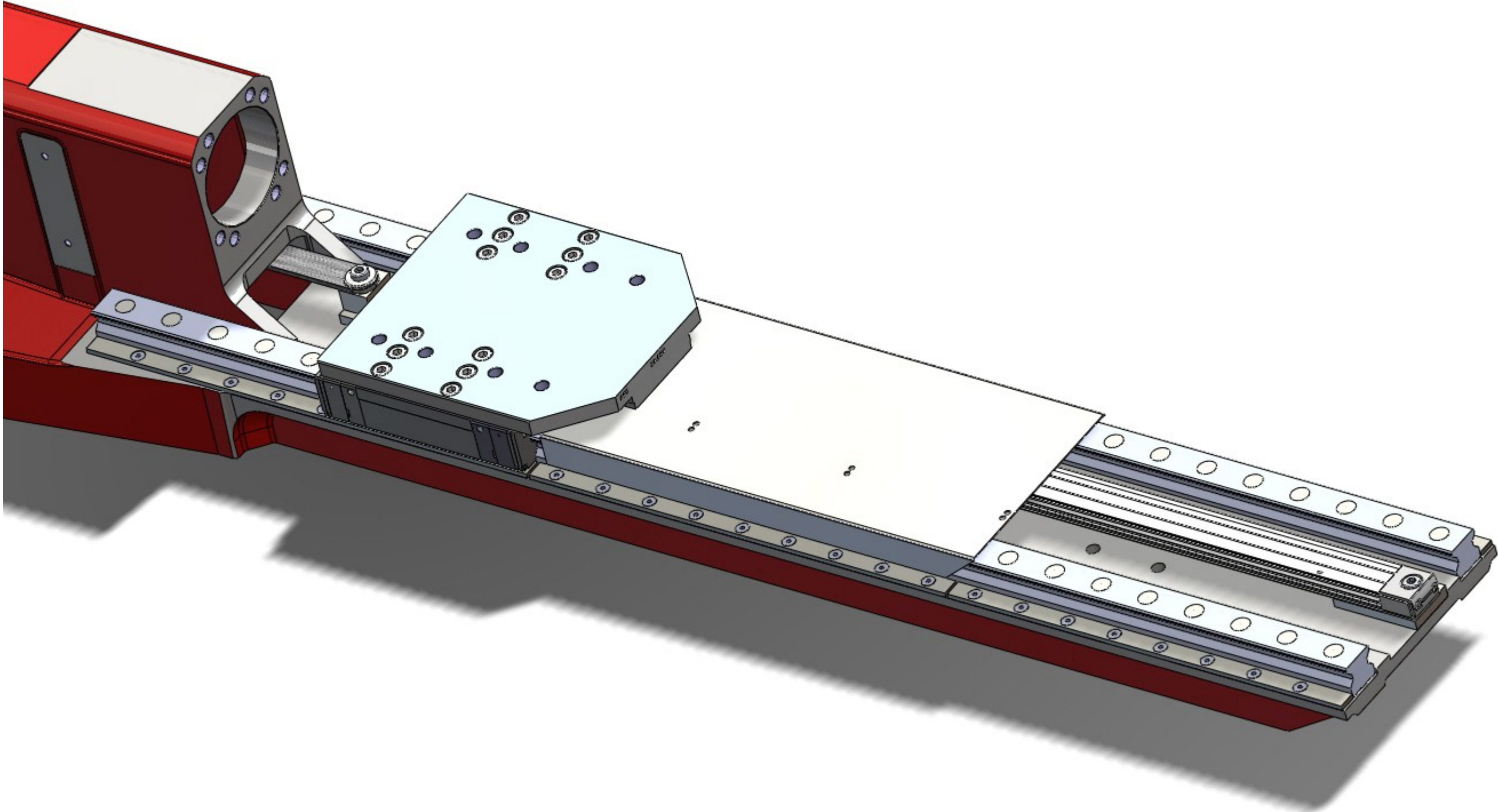
DRILL

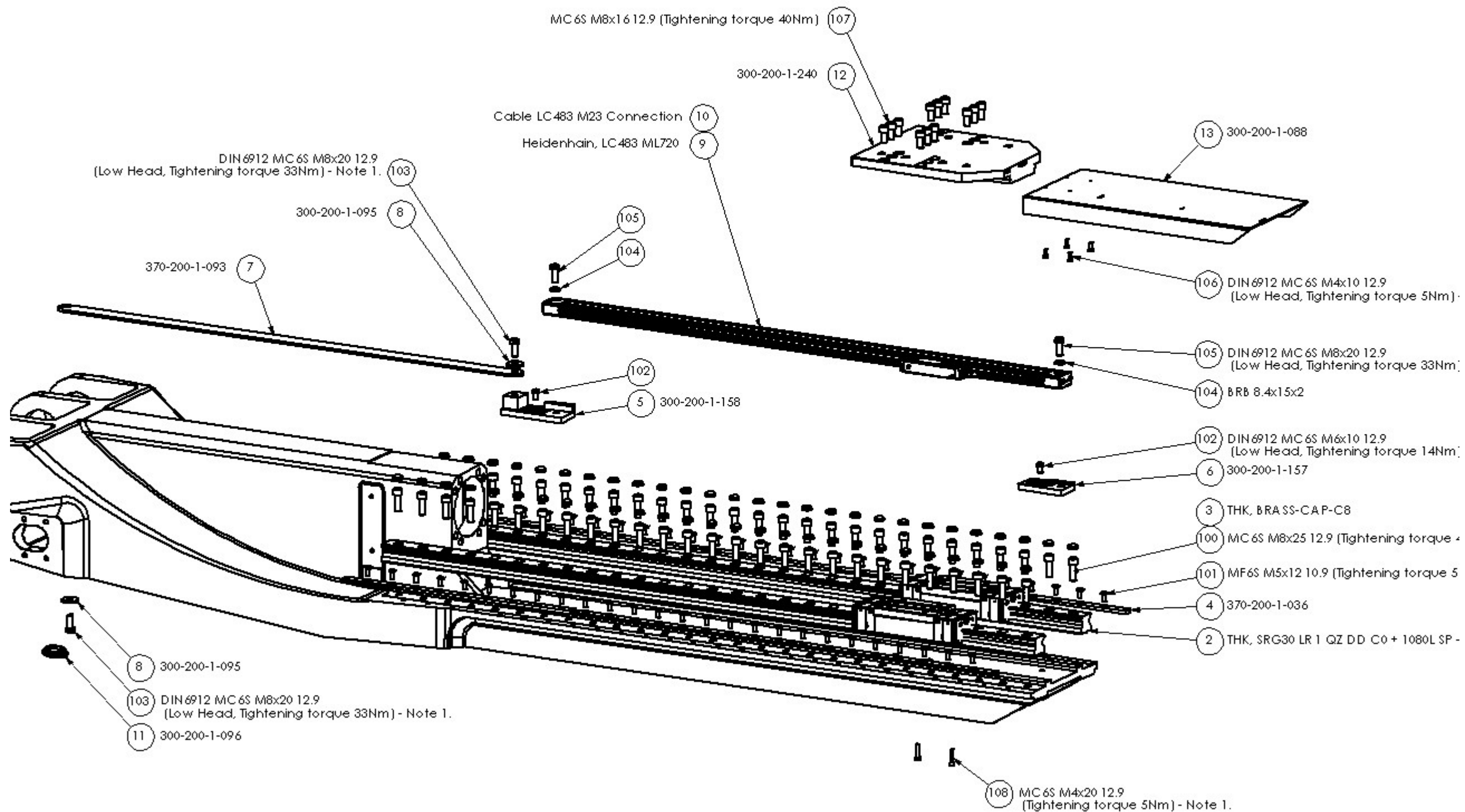
XT 700S (PKM)

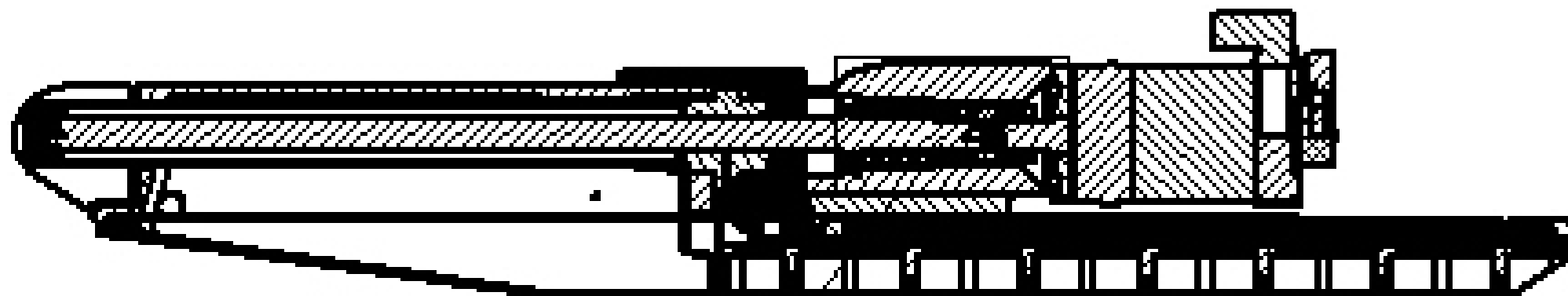


Machine Specification/Features :

- Siemens 840D SL CNC System with Exechon Transformation
- Number of Linear axes : 3 (X, Y, Z) with Absolute Linear Encoders (Heidenhain)
- Linear Axes Motors : Water Cooled, 43Nm, 3000RPM (Siemens)
- Linear Axes Stroke : 700mm
- Roller Screw for the Linear Axes : 36mm x 10mm (Rollvis)
- Number of Rotary axes : 2 (C, A) with Hollow Shaft Incremental Angle Encoders (Heidenhain)
- Rotary Axes Motors : Natural Cooled, 20Nm, 15Nm, 2000RPM (Siemens)
- Rotary Axes Strokes : $\pm 360^\circ$ and 140°
- Harmonic Drive Gearbox with Timing Belt for Rotary Axes
- Integrated Spindle (Weiss)
- Spindle RPM : 18,000RPM
- Spindle Power : 32.5kW
- Spindle taper : HSK A63
- Spindle Size : 150mm
- Work Envelope : 2000mm
- Acceleration : 2g
- Positioning Accuracy : $10\mu\text{m}$
- Repeatability : $5\mu\text{m}$
- Approximate Weight of the m/c without frame : 1800kg







SECTION A-A

